Assignment no : 01

Perform the following operations using Python on any open-source dataset (e.g., data.csv)

1. **Import all the required Python Libraries.**
2. **Locate an open-source data from the web (e.g. https://www.kaggle.com). Provide a clear description of the data and its source (i.e., URL of the web site).**
3. **Load the Dataset into pandas' data frame.**
4. **Data Preprocessing: check for missing values in the data using pandas isnull (), describe function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame.**
5. **Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.**
6. **Turn categorical variables into quantitative variables in Python. In addition to the codes and outputs, explain every operation that you do in the above steps and explain everything that you do to import/read/scrape the data set.**

In [47]:

```
!pip3 install pandas
!pip3 install numpy
!pip3 install matplotlib
!pip3 install seaborn
```

```
Requirement already satisfied: pandas in /Library/Frameworks/Python.framework/Versions/3.
12/lib/python3.12/site-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /Library/Frameworks/Python.framework/Vers
ions/3.12/lib/python3.12/site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in /Library/Frameworks/Python.frame
work/Versions/3.12/lib/python3.12/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Library/Frameworks/Python.framework/Versi
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Requirement already satisfied: tzdata>=2022.7 in /Library/Frameworks/Python.framework/Ver
sions/3.12/lib/python3.12/site-packages (from pandas) (2024.1)
Requirement already satisfied: six>=1.5 in /Library/Frameworks/Python.framework/Versions/
3.12/lib/python3.12/site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: numpy in /Library/Frameworks/Python.framework/Versions/3.1
2/lib/python3.12/site-packages (1.26.4)
Requirement already satisfied: matplotlib in /Library/Frameworks/Python.framework/Version
s/3.12/lib/python3.12/site-packages (3.8.4)
Requirement already satisfied: contourpy>=1.0.1 in /Library/Frameworks/Python.framework/V
ersions/3.12/lib/python3.12/site-packages (from matplotlib) (1.2.1)
Requirement already satisfied: cycler>=0.10 in /Library/Frameworks/Python.framework/Versi
ons/3.12/lib/python3.12/site-packages (from matplotlib) (0.12.1)
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Requirement already satisfied: pillow>=8 in /Library/Frameworks/Python.framework/Versions
/3.12/lib/python3.12/site-packages (from matplotlib) (10.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /Library/Frameworks/Python.framework/V
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Requirement already satisfied: python-dateutil>=2.7 in /Library/Frameworks/Python.framewo
rk/Versions/3.12/lib/python3.12/site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in /Library/Frameworks/Python.framework/Versions/
3.12/lib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib) (1.16.0)
Requirement already satisfied: seaborn in /Library/Frameworks/Python.framework/Versions/3
.12/lib/python3.12/site-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /Library/Frameworks/Python.framewo
```

```
rk/Versions/3.12/lib/python3.12/site-packages (from seaborn) (1.26.4)
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Requirement already satisfied: fonttools>=4.22.0 in /Library/Frameworks/Python.framework/
Versions/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (4.51.
0)
Requirement already satisfied: kiwisolver>=1.3.1 in /Library/Frameworks/Python.framework/
Versions/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.4.5
)
Requirement already satisfied: packaging>=20.0 in /Library/Frameworks/Python.framework/Ve
rsions/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (24.0)
Requirement already satisfied: pillow>=8 in /Library/Frameworks/Python.framework/Versions
/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (10.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /Library/Frameworks/Python.framework/V
ersions/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (3.1.2)
Requirement already satisfied: python-dateutil>=2.7 in /Library/Frameworks/Python.framewo
rk/Versions/3.12/lib/python3.12/site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (2.
9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /Library/Frameworks/Python.framework/Versi
ons/3.12/lib/python3.12/site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in /Library/Frameworks/Python.framework/Ver
sions/3.12/lib/python3.12/site-packages (from pandas>=1.2->seaborn) (2024.1)
Requirement already satisfied: six>=1.5 in /Library/Frameworks/Python.framework/Versions/
3.12/lib/python3.12/site-packages (from python-dateutil>=2.7->matplotlib!=3.6.1,>=3.4->se
aborn) (1.16.0)
```

Part 1

Data Wrangling Data wrangling, also known as data munging, is the process of cleaning, transforming, and organizing raw data into a structured format suitable for analysis.

1. **import the required libraries**

In [48]:

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the dataset using pandas library

In [49]:

```python
data = pd.read_csv('data_titanic.csv')
```

Check the contents of dataset using df.head() and df.tail() functions

In [50]:

```python
data.head(5)
```

Out[50]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence	female	38.0	1	0	PC 17599	71.2833	C85	C

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
				Briggs Th								
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

In [51]:

```
data.tail()
```

Out[51]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

Print the name of all columns of the data

In [52]:

```
data.columns
```

Out[52]:

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='object')
```

In [53]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [54]:

```
data.describe()
```

Out[54]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

count the values

In [55]:

```
data['Age'].value_counts()
```

Out[55]:

```
Age
24.00    30
22.00    27
18.00    26
19.00    25
28.00    25
         ..
36.50     1
55.50     1
0.92      1
23.50     1
74.00     1
Name: count, Length: 88, dtype: int64
```

In [56]:

```
data[['Survived','Pclass']].value_counts()
```

Out[56]:

```
Survived  Pclass
0         3         372
1         1         136
          3         119
0         2          97
1         2          87
0         1          80
Name: count, dtype: int64
```

Evaluating for NULL Data **The missing values are converted to Python's default. We use Python's built-in functions to identify these missing values. There are two methods to detect missing data:**

- **isnull()**
- **.notnull() The output is a boolean value indicating whether the value that is passed into the argument is in fact missing data. "True" stands for missing value, while "False" stands for not missing value.**

Deal with missing data

1. Drop data

- **Drop the whole row**
- **Drop the whole column**

2.Replace data

- **Replace it by mean**

- **Replace it by frequency / mode**
- **Replace it based on other functions**

In [57]:

```
data.isnull()
```

Out[57]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	False	False	False	False	False	False	False	False	False	False	True	False
1	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	True	False
3	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	True	False
...	...	...	...	...	...	...	...	...	...	...	...	...
886	False	False	False	False	False	False	False	False	False	False	True	False
887	False	False	False	False	False	False	False	False	False	False	False	False
888	False	False	False	False	False	True	False	False	False	False	True	False
889	False	False	False	False	False	False	False	False	False	False	False	False
890	False	False	False	False	False	False	False	False	False	False	True	False

891 rows × 12 columns

In [58]:

```
data.isnull().sum()
```

Out[58]:

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [59]:

```
data_missing = data.isna().sum()
```

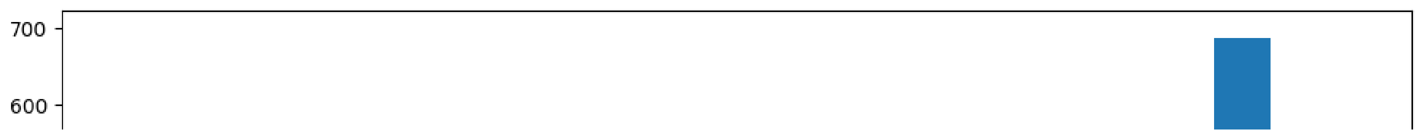
The `figsize` parameter sets the size of the figure, and `rot` controls the rotation of x-axis labels (0 means horizontal).

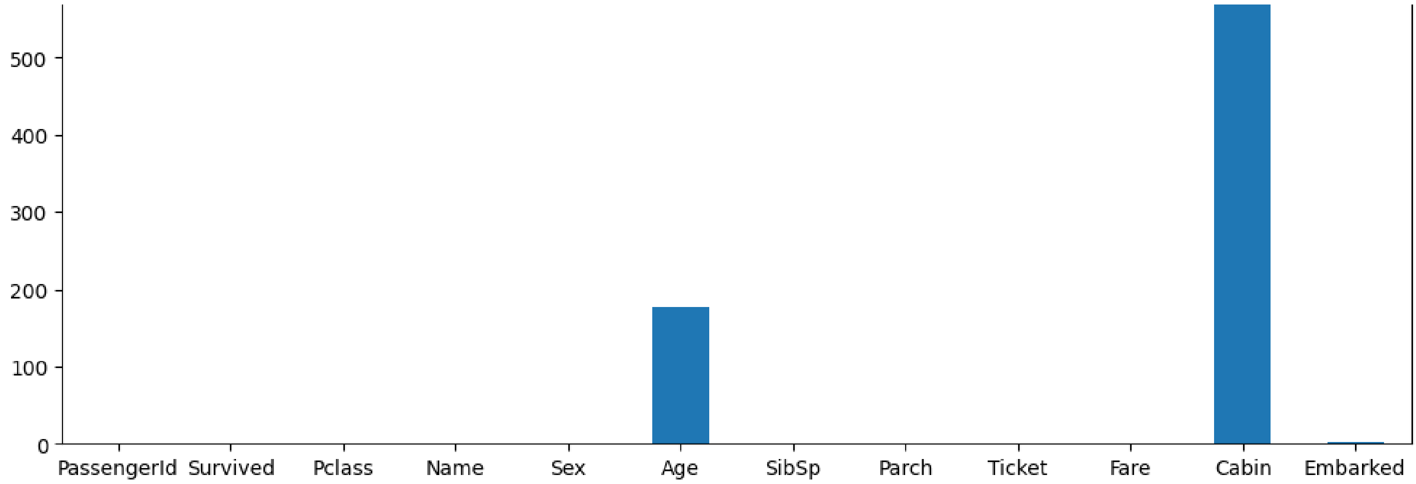
In [60]:

```
data_missing.plot.bar(figsize=(12,5),rot=0)
```

Out[60]:

```
<Axes: >
```





In [61]:

```
data.notnull().sum()
```

Out[61]:

```
PassengerId    891
Survived       891
Pclass         891
Name           891
Sex            891
Age            714
SibSp          891
Parch          891
Ticket         891
Fare           891
Cabin          204
Embarked       889
dtype: int64
```

Dimension of DataFrame

In [62]:

```
data.shape
```

Out[62]:

```
(891, 12)
```

Fill the null values

In [63]:

```
mean_age = data['Age'].mean()
data['Age'] = data['Age'].fillna(value = mean_age)
print(data['Age'].isnull().sum())
```

```
0
```

In [64]:

```
variable_descriptions = data.info()
variable_types = data.dtypes
print("Variable descriptions:\n", variable_descriptions)
print("\nVariable types:\n", variable_types)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
```

```
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          891 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
Variable descriptions:
 None

Variable types:
 PassengerId      int64
Survived         int64
Pclass           int64
Name            object
Sex             object
Age            float64
SibSp            int64
Parch            int64
Ticket          object
Fare           float64
Cabin           object
Embarked        object
dtype: object
```

5. Data Normalization

In [65]:

```
data['Age'] = pd.to_numeric(data['Age'], errors='coerce')
data['Age']
```

Out[65]:

```
0      22.000000
1      38.000000
2      26.000000
3      35.000000
4      35.000000
         ...
886    27.000000
887    19.000000
888    29.699118
889    26.000000
890    32.000000
Name: Age, Length: 891, dtype: float64
```

Assigning Data Tyes

In [66]:

```
data['PassengerId'] = data['PassengerId'].astype(int)
data['Survived'] = data['Survived'].astype(int)
data['Pclass'] = data['Pclass'].astype(int)
data['SibSp'] = data['SibSp'].astype(int)
data['Parch'] = data['Parch'].astype(int)
# nvert Sex and Embarked to category
data['Sex'] = data['Sex'].astype('category')
data['Embarked'] = data['Embarked'].astype('category')

# After making the necessary changes, check the structure of the data again
print("\nDataset after Data Type Conversion:")
print(data.dtypes)
```

```
Dataset after Data Type Conversion:
```

```
Dataset after Data Type Conversion:
PassengerId        int64
Survived           int64
Pclass             int64
Name              object
Sex             category
Age              float64
SibSp              int64
Parch              int64
Ticket            object
Fare             float64
Cabin             object
Embarked        category
dtype: object
```

In [67]:

```
!pip3 install scikit-learn
```

```
Requirement already satisfied: scikit-learn in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (1.4.2)
Requirement already satisfied: numpy>=1.19.5 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from scikit-learn) (1.26.4)
Requirement already satisfied: scipy>=1.6.0 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from scikit-learn) (1.13.0)
Requirement already satisfied: joblib>=1.2.0 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from scikit-learn) (1.4.0)
Requirement already satisfied: threadpoolctl>=2.0.0 in /Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-packages (from scikit-learn) (3.5.0)
```

LabelEncoder: Encodes categorical labels into numerical values.
data['Sex'] = le.fit_transform(data['Sex']): Applies the LabelEncoder to the 'Sex' column in the DataFrame 'data', transforming categorical values into numerical labels.

In [68]:

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

data['Sex'] = le.fit_transform(data['Sex'])
data.head()
```

Out[68]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	NaN	S

Standardization

- **StandardScaler: Scikit-learn's built-in class for standard scaling.**
- **scaler.fit_transform(data['Fare'].values.reshape(-1,1)): Fits the StandardScaler to the 'Fare' column in the DataFrame 'data' and transforms it, reshaping the data to ensure it's a 2D array.**

In [69]:

```python
import numpy as np

def standard_scaler(data):
    mean = np.mean(data)
    std_dev = np.std(data)
    scaled_data = [(x-mean) / std_dev for x in data]
    return scaled_data

from sklearn.preprocessing import StandardScaler
standardized_data = scaler.fit_transform(data['Fare'].values.reshape(-1,1))

print(standardized_data.mean())
```

```
0.06285842768394742
```

Normalization

In [70]:

```python
from sklearn.preprocessing import MinMaxScaler

def minMaxScaler(data):
    min_val = min(data)
    max_val = max(data)
    scaled_data = [(x-min_val)/(max_val-min_val) for x in data]
    return scaled_data


scaler = MinMaxScaler()
normalized_data = scaler.fit_transform(data['Fare'].values.reshape(-1, 1))
print(normalized_data)
```

```
[[0.01415106]
 [0.13913574]
 [0.01546857]
 [0.1036443 ]
 [0.01571255]
 [0.0165095 ]
 [0.10122886]
 [0.04113566]
 [0.02173075]
 [0.05869429]
 [0.03259623]
 [0.05182215]
 [0.01571255]
 [0.06104473]
 [0.01533038]
 [0.03122992]
 [0.05684821]
 [0.02537431]
 [0.03513366]
 [0.01410226]
 [0.05074862]
 [0.02537431]
 [0.01567195]
 [0.06929139]
 [0.04113566]
 [0.06126432]
 [0.01410226]
 [0.51334181]
 [0.01537917]
 [0.01541158]
 [0.0541074 ]
 [0.28598956]
 [0.01512699]
 [0.02049464]
 [0.16038672]
 [0.10149724]
 [0.01411046]
 [0.01571255]
 [0.03513366]
 [0.02194234]
```

```
[0.01849397]
[0.04098927]
[0.01541158]
[0.08115719]
[0.01537917]
[0.01571255]
[0.03025399]
[0.01512699]
[0.04231498]
[0.03474329]
[0.07746484]
[0.01522459]
[0.14976542]
[0.05074862]
[0.12097534]
[0.06929139]
[0.02049464]
[0.01411046]
[0.05416439]
[0.0915427 ]
[0.01411046]
[0.1561496 ]
[0.16293235]
[0.05445717]
[0.0541074 ]
[0.02975782]
[0.02049464]
[0.01592394]
[0.01546857]
[0.01690807]
[0.02049464]
[0.0915427 ]
[0.14346245]
[0.02821272]
[0.11027246]
[0.01493181]
[0.01541158]
[0.01571255]
[0.05660423]
[0.02434958]
[0.01756683]
[0.01854277]
[0.01520019]
[0.09193308]
[0.02049464]
[0.03093714]
[0.06709553]
[0.01571255]
[0.51334181]
[0.01571255]
[0.01571255]
[0.01533038]
[0.11940565]
[0.04015973]
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